Introduction to Natural Language Processing

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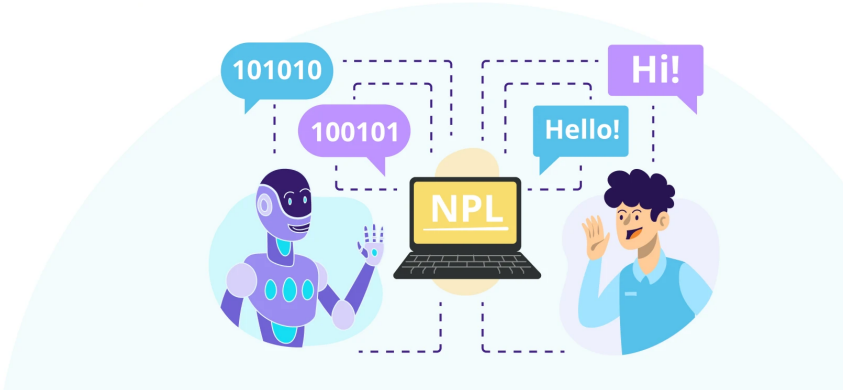
Natural Language Processing



Source: Byteridge, "Natural Language Processing Business Use Cases".

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Motivation

- Language helps us talk to each other, and now to computers too.
- NLP (Natural Language Processing) teaches machines to understand, interpret, and generate human language.
- Why is NLP important?
  - Makes **chatbots** like ChatGPT and Siri possible
  - Powers **search engines** like Google
  - Helps **translate languages** (Google Translate)
  - Finds out what **people feel in reviews** (sentiment analysis)
- NLP is foundational to advanced AI systems.

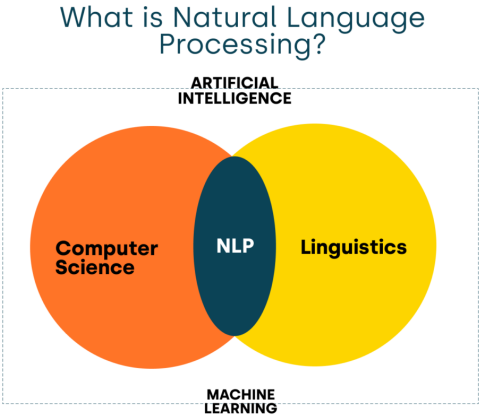
Learning Outcomes

- **Define** Natural Language Processing (NLP) and describe its core tasks
- **Apply** classical text classification with Naive Bayes and logistic regression
- **Explain and construct** N-grams (unigram, bigram, trigram) using sequence notation
- **Compute** N-gram probabilities, count matrices, and sequence probabilities
- **Apply** start/end tokens and handle unknown words (OOV, UNK)
- **Smooth** N-gram estimates so unseen contexts receive non-zero probability
- **Evaluate** a language model with perplexity
- **Recognize** the limitations of N-gram models and future directions

Natural Language Processing: **Introduction**

What is NLP?

- Study of computational approaches to processing natural languages.
- Processing includes:
  - Acquiring language data
  - Representing information
  - Storing text and speech
  - Understanding meaning
  - Characterizing language patterns
  - Generating new language
- Natural languages refer to human languages.



Source: Qtravel.ai, "What is Natural Language Processing".

NLP - Introduction (cont.)

NLP = Computer Science + Linguistics + AI

- **Deals with:**
  - Language understanding (input)
  - Language generation (output)
  - Language translation
  - Information extraction
- **Subfields:**
  - Syntax
  - Semantics
  - Pragmatics
  - Discourse

Goal

Goal: Deep Understanding
Requires context, linguistic structure, meanings...

To Avoid: Shallow Matching
Could be useful also though depending on use case

Cartoon: Gary Larson, *The Far Side*.

Goal of NLP:

- Enable machines to understand and generate human language.
- Facilitate human-computer interaction through natural language.
- Develop systems that can process and analyze large amounts of text data.

NLP Pipeline

- **Text Preprocessing**
  - Cleaning and preparing raw text for analysis.
- **Tokenization**
  - Splitting text into words, sentences, or other meaningful units.
- **POS Tagging**
  - Assigning parts of speech (noun, verb, etc.) to each token.
- **Parsing**
  - Analyzing grammatical structure of sentences.
- **Named Entity Recognition (NER)**
  - Identifying entities such as people, organizations, locations.
- **Sentiment Analysis / Classification**
  - Determining sentiment or categorizing text.
- **Language Modeling**
  - Predicting the next word or sequence in text.

Text Data is Superficial

- An iceberg is a large piece of freshwater ice that has broken off from a snow-formed glacier or ice shelf and is floating in open water.

Source: Dan Klein, UC Berkeley NLP slides (via T. Berg-Kirkpatrick, CMU).

Text Data is Superficial

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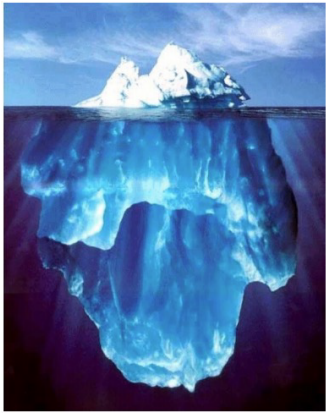


Source: Dan Klein, UC Berkeley NLP slides.

Core NLP Tasks

Task	What It Does	Example
Tokenization	Split text into words	`"I love NLP"` → `["I", "love", "NLP"]`
POS Tagging	Label grammar tags	`"Dogs bark"` → [Noun, Verb]
Named Entity Recognition	Find names, places, etc.	`"Christopher Nolan lives in Los Angeles"` → [Person, Location]
Sentiment Analysis	Detect mood	`"This movie was amazing!"` → Positive
Machine Translation	Language to language	English → French

Applications such as summarization and question answering chain several of these tasks together.

Text Preprocessing

- **Lowercase everything**: `"NLP"` → `"nlp"`
- **Removing stop words**: drop words that carry little meaning on their own, such as “is”, “the” and “and”.
- **Stemming / Lemmatization**: reduce a word to its root or base form, `"running"` → `"run"`
- **Language detection** (optional): identify each document’s language before applying language-specific steps.
- **Spellcheck** (optional): normalize typos so variants map to the same token.
- **Vectorization**: turn the surviving tokens into numbers a model can consume.

Corpora

The Rosetta Stone, British Museum: one text, three parallel scripts. Source: Dan Klein, UC Berkeley NLP slides.

What is a corpus?

- A corpus is a collection of text.
- Often annotated in some way.
- Sometimes just lots of text.
- **Balanced corpora:** Usually not possible in practice.
- **Examples:**
  - Newswire collections: 500M+ words
  - Brown corpus: 1M words of tagged "balanced" text
  - Penn Treebank: 1M words of parsed WSJ
  - Canadian Hansards: 10M+ words of aligned French/English sentences
  - The Web: billions of words of who knows what

What is Vocabulary in NLP?

- **Vocabulary** = All unique words in your dataset
- Example: `"I love NLP and NLP loves me"` → Vocabulary = {`"I"`, `"love"`, `"NLP"`, `"and"`, `"loves"`, `"me"`}
- More data = Bigger vocabulary = Harder to process!

Tip: Rare words may not help; common words may not mean much.

The Problem with Sparse Representations

Traditional approach: One-hot encoding

- Example: `"NLP"` $\rightarrow$ [0, 0, 1, 0, 0, 0, 0...]

Problem:

- High-dimensional (thousands of words!)
- Sparse (mostly 0s)
- No meaning in structure (no relation between `"king"` and `"queen"`)
- ✖ Not efficient for learning

Feature extraction = Turning text into numbers

- Count how often each word appears (**Term Frequency**)

Example:

Text	`"great product"`	`"bad product"`
Word: `"great"`	1	0
Word: `"bad"`	0	1

⚖ Use this to find patterns in sentiment, spam, etc.

Positive & Negative Frequencies

Suppose you're classifying reviews:

Positive Reviews: `["amazing", "good", "great"]`
Negative Reviews: `["bad", "awful", "terrible"]`

Count how often each word appears in each class.

Example table:

Word	Positive Count	Negative Count
good	20	1
bad	1	30

Helps models detect the "tone" (**sentiment clues**) of new text.

Feature Extraction Techniques

- **Bag of Words (BoW)**: Just counts word frequencies in each document.
- **TF-IDF (Term Frequency-Inverse Document Frequency)**: Adjusts for how "unique" or important a word is in a document compared to all documents.
  - Words like `"the"`, `"is"` are less important.
- **Word Embeddings** (Word2Vec and successors, covered later): learn dense vectors that place related words near each other, so similarity and analogy become geometry rather than string matching.

BoW and TF-IDF both throw word order away. That is enough for topic and sentiment, but not for anything that depends on which word follows which.

NLP: **Classical Text Classification**

Bayes' Rule – A Quick Look

Bayes' Rule:

$$P(\text{Class} \mid \text{Words}) = \frac{P(\text{Words} \mid \text{Class}) \cdot P(\text{Class})}{P(\text{Words})}$$

- Predict class (e.g., spam or not) given the words
- Simple, powerful tool used in Naive Bayes classifiers
- Helps models reason with uncertainty

Naive Bayes for Sentiment Analysis

Simple, fast probabilistic model

Assumes:

- Words are independent given the class (a "naive" assumption)

Example:

- `"This movie is awesome"` $\rightarrow P(\text{Positive})$ high
- `"This movie is boring"` $\rightarrow P(\text{Negative})$ high

✔ Works well with BoW, TF-IDF, and N-grams

Logistic Regression for Text Classification

Logistic regression is a probability-based classifier we have already met; only the features change here.

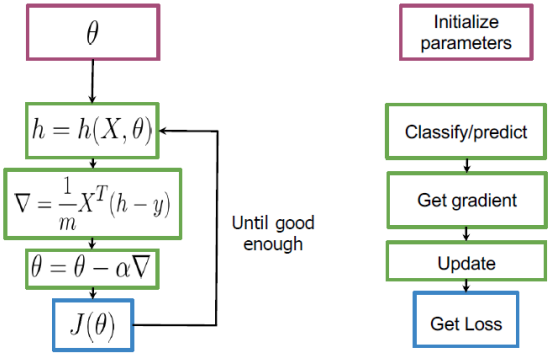
- It takes word features (from BoW or TF-IDF)
- It learns weights to predict if text belongs to a certain class, such as:
  - Positive or Negative
  - Spam or Not Spam
- Outputs a probability:

$$P(\text{Class} = 1 \mid \text{features}) = \sigma(w_1 x_1 + w_2 x_2 + \ldots + b)$$

- Words like "awesome" get high weights for positive class

Intuition: each word casts a weighted "vote" toward a class, and training decides how loud each vote is.

Logistic Regression for Text Classification (cont.)



Training a Logistic Regression Model. Source: DeepLearning.AI, *NLP with Classification and Vector Spaces*, Week 1 slides.

Natural Language Processing: **N-grams**

What are N-grams?

An N-gram is a sequence of N words

Corpus: I am happy because I am learning

Unigrams: { I , am , happy , because , learning }

Bigrams: { I am , am happy , happy because ... }

Trigrams: { I am happy , am happy because, ... }

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

- Items are typically **words** or **characters**.

Tokenization (Basics)

- Tokenization is the process of breaking text into smaller units called **tokens**.
- Tokens can be words, characters, or subwords.
- Example: "I love NLP" can be tokenized into:
  - Words: `["I", "love", "NLP"]`
  - Characters: `["I", " ", "l", "o", "v", "e", " ", "N", "L", "P"]`
- Subword tokenizers sit between the two: frequent words stay whole, rare or long ones are split into pieces the model already knows, so `"tokenization"` becomes `["token", "ization"]`.
- Every later step counts tokens, so the choice of tokenizer fixes what the model can and cannot see.

Examples of N-grams

Sentence: "I love NLP"

- **Unigrams:** I, love, NLP
- **Bigrams:** I love, love NLP
- **Trigrams:** I love NLP

Why Use N-grams?

- Capture local word co-occurrence
- Build simple language models
- Easy to compute and analyze
- Trade-off between simplicity (unigram) and contextual richness (trigram)

Sequence Notation Basics

Sentence: $w_1, w_2, \ldots, w_n$

For example: $w_1 = \text{I},\ w_2 = \text{love},\ w_3 = \text{NLP}$

General representation:

- Unigram: $P(w_i)$
- Bigram: $P(w_i \mid w_{i-1})$
- Trigram: $P(w_i \mid w_{i-2},\ w_{i-1})$

Sequence Notation Example:

Corpus: This is great ... teacher drinks tea.

$w_1 \; w_2 \; w_3 \quad \quad w_{498} \; w_{499} \; w_{500}$

$m = 500$

$$w_1^m = w_1 \; w_2 \; \ldots \; w_m$$

$$w_1^3 = w_1 \; w_2 \; w_3$$

$$w_{m-2}^m = w_{m-2} \; w_{m-1} \; w_m$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Unigram probability

Corpus: I am happy because I am learning

Size of corpus m = 7

$$P(I) = \frac{2}{7}$$

$$P(happy) = \frac{1}{7}$$

Probability of unigram: $$P(w) = \frac{C(w)}{m}$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Bigram probability

Corpus: I am happy becaus e am learning
I

$$P(am|I) = \frac{C(I\ am)}{C(I)} = \frac{2}{2} = 1$$

$$P(happy|I) = \frac{C(I\ happy)}{C(I)} = \frac{0}{2} = 0$$

I happy

$$P(learning|am) = \frac{C(am\ learning)}{C(am)} = \frac{1}{2}$$

Probability of a bigram: $P(y|x) = \frac{C(x\ y)}{\sum_w C(x\ w)} = \frac{C(x\ y)}{C(x)}$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Trigram probability

Corpus: I am happy because I am learning

$$P(happy|I\ am) = \frac{C(I\ am\ happy)}{C(I\ am)} = \frac{1}{2}$$

Probability of a trigram: $$P(w_3|w_1^2) = \frac{C(w_1^2\ w_3)}{C(w_1^2)}$$

$$C(w_1^2\ w_3) = C(w_1\ w_2\ w_3) = C(w_1^3)$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

N-gram probability

Probability of N-gram: $$P(w_N|w_1^{N-1}) = \frac{C(w_1^{N-1}\, w_N)}{C(w_1^{N-1})}$$

$$C(w_1^{N-1}\, w_N) = C(w_1^N)$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Objective: Compute the probability of a sentence

N-gram Assumption: The probability of a word depends only on the previous $(n-1)$ words.

Formula:

$$P(w_1^n) \approx \prod_{i=1}^{n} P(w_i \mid w_{i-n+1}^{i-1})$$

where w_1^n denotes the sequence $w_1, w_2, \ldots, w_n$ and w_{i-n+1}^{i-1} is the context of the previous $(n-1)$ words.

Bigram MLE:

$$P(w_i \mid w_{i-1}) = \frac{\text{Count}(w_{i-1}, w_i)}{\text{Count}(w_{i-1})}$$

- $\text{Count}(w_{i-1}, w_i)$: Number of times the bigram (w_{i-1}, w_i) appears in the corpus.
- $\text{Count}(w_{i-1})$: Number of times the word w_{i-1} appears as a context.

Example: Bigram Probabilities

Text: "I love NLP. I love AI."

Bigrams and Counts:

- (I, love): 2
- (love, NLP): 1
- (love, AI): 1

Probability Calculations:

- $P(\text{love} \mid \text{I}) = \frac{2}{2} = 1.0$
- $P(\text{NLP} \mid \text{love}) = \frac{1}{2} = 0.5$
- $P(\text{AI} \mid \text{love}) = \frac{1}{2} = 0.5$

Probability of a sequence

- Given a sentence, what is its probability?

$$P(the\ teacher\ drinks\ tea) = ?$$

- Conditional probability and chain rule reminder

$$P(B|A) = \frac{P(A,B)}{P(A)} \implies P(A,B) = P(A)P(B|A)$$

$$P(A,B,C,D) = P(A)P(B|A)P(C|A,B)P(D|A,B,C)$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Probability of a sequence

$$P(the\ teacher\ drinks\ tea) =$$

$$P(the)P(teacher|the)P(drinks|the\ teacher)$$
$$P(tea|the\ teacher\ drinks)$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Sentence not in corpus

Problem: Corpus almost never contains the exact sentence we are interested in or even its longer subsequences.

Input: the teacher drinks tea

$$P(the\ teacher\ drinks\ tea) = P(the)P(teacher|the)P(drinks|the\ teacher)P(tea|the\ teacher\ drinks)$$

$$P(tea|the\ teacher\ drinks) = \frac{C(the\ teacher\ drinks\ tea)}{C(the\ teacher\ drinks)}$$

Both likely 0

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Approximation of sequence probability

the teacher drinks tea

$$P(tea|the\ teacher\ drinks) \approx P(tea|drinks)$$

$$P(teacher|the)$$
$$P(drinks|teacher)$$
$$P(tea|drinks)$$

$$P(the\ teacher\ drinks\ tea) =$$

$$P(the)P(teacher|the)P(drinks|the\ teacher)P(tea|the\ teacher\ drinks)$$

$$P(the)P(teacher|the)P(drinks|teacher)P(tea|drinks)$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Approximation of sequence probability

- Markov assumption: only last N words matter

- Bigram $P(w_n|w_1^{n-1}) \approx P(w_n|w_{n-1})$
- N-gram $P(w_n|w_1^{n-1}) \approx P(w_n|w_{n-N+1}^{n-1})$

- Entire sentence modeled with bigram $P(w_1^n) \approx \prod_{i=1}^{n} P(w_i|w_{i-1})$

$$P(w_1^n) \approx P(w_1)P(w_2|w_1)...P(w_n|w_{n-1})$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Quiz

Objective: Apply sequence probability approximation with bigrams.

Question:
Given these conditional probabilities
P(Mary)=0.1; P(likes)=0.2; P(cats)=0.3
P(Mary|likes) =0.2; P(likes|Mary) =0.3; P(cats|likes)=0.1; P(likes|cats)=0.4

Approximate the probability of the following sentence with bigrams: "Mary likes cats"

Type: Multiple Choice, single answer

Options and solution:

1. P(Mary likes cats) = 0
2. P(Mary likes cats) =1
3. P(Mary likes cats) = 0.003
4. P(Mary likes cats) = 0.008

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

N-gram Models: **Start and End Tokens**

Why Use Special Tokens?

- **Start:** <s> or <start>
- **End:** </s> or <end>

Benefits:

- Helps model sentence boundaries
- Enables generation and evaluation

Start of sentence token <s>

the teacher drinks tea

$$P(the\ teacher\ drinks\ tea) \approx P(the)P(teacher|the)P(drinks|teacher)P(tea|drinks)$$

<s> the teacher drinks tea

$$P(<s>\ the\ teacher\ drinks\ tea) \approx P(the|<s>)P(teacher|the)P(drinks|teacher)P(tea|drinks)$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Start of sentence token <s> for N-grams

- Trigram:

$$P(the\ teacher\ drinks\ tea) \approx P(the)P(teacher|the)P(drinks|the\ teacher)P(tea|teacher\ drinks)$$

the teacher drinks tea => <s> <s> the teacher drinks tea

$$P(w_1^n) \approx P(w_1|<s>\ <s>)P(w_2|<s>\ w_1)...P(w_n|w_{n-2}\ w_{n-1})$$

- N-gram model: add N-1 start tokens <s>

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

End of sentence token </s> - motivation

$$P(y|x) = \frac{C(x\ y)}{\sum_w C(x\ w)} = \frac{C(x\ y)}{C(x)}$$

Corpus:

<s> Lyn drinks chocolate

<s> John drinks

$$\sum_w C(drinks\ w) = 1$$

$$C(drinks) = 2$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

End of sentence token </s> - motivation (cont.)

Corpus

<s> yes no

<s> yes yes

<s> no no

Sentences of length 2:

<s> yes yes

<s> yes no

<s> no no

<s> no yes

$$P(< s > \text{ yes yes}) =$$

$$P(\text{yes} \mid < s >) \times P(\text{yes} \mid \text{yes}) =$$

$$\frac{C(< s > \text{ yes})}{\sum_w C(< s > w)} \times \frac{C(\text{yes yes})}{\sum_w C(\text{yes } w)} =$$

$$\frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

End of sentence token </s> - motivation (cont.)

Corpus

<s> yes no

<s> yes yes

<s> no no

Sentences of length 2:

<s> yes yes

<s> yes no

<s> no no

<s> no yes

$$P(< s > \text{ yes yes}) = \frac{1}{3}$$

$$P(< s > \text{ yes no}) = \frac{1}{3}$$

$$P(< s > \text{ no no}) = \frac{1}{3}$$

$$P(< s > \text{ no yes}) = 0$$

$$\sum_{\text{2 word}} P(\cdots) = 1$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

End of sentence token </s> - motivation (cont.)

Corpus

<s> yes no

<s> yes yes

<s> no no

Sentences of length 3:

<s> yes yes yes

<s> yes yes no

...

<s> no no no

$$P(< s > \text{ yes yes yes}) = \dots$$

$$P(< s > \text{ yes yes no}) = \dots$$

$$\dots = \dots$$

$$P(< s > \text{ no no no}) = \dots$$

$$\sum_{3 \text{ word}} P(\cdots) = 1$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

End of sentence token </s> - motivation (cont.)

Corpus

<s> yes no

<s> yes yes

<s> no no

$$\sum_{2 \text{ word}} P(\cdots) + \sum_{3 \text{ word}} P(\cdots) + \ldots = 1$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

End of sentence token </s> - solution

- Bigram

<s> the teacher drinks tea => <s> the teacher drinks tea </s>

$$P(the|<s>)P(teacher|the)P(drinks|teacher)P(tea|drinks)P(</s>|tea)$$

Corpus:

<s> Lyn drinks chocolate </s>

<s> John drinks </s>

$$\sum_{w} C(drinks\ w) = 2$$

$$C(drinks) = 2$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

End of sentence token </s> for N-grams

- N-gram => just one </s>

E.g. Trigram:

the teacher drinks tea => <s> <s> the teacher drinks tea </s>

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Example - bigram

Corpus:

- <s> Lyn drinks chocolate </s>
- <s> John drinks tea </s>
- <s> Lyn eats chocolate </s>

$$P(\text{Lyn} \mid \text{<s>}) = \frac{2}{3} \qquad P(\text{John} \mid \text{<s>}) = \frac{1}{3}$$

$$P(\text{drinks} \mid \text{Lyn}) = \frac{1}{2} \qquad P(\text{chocolate} \mid \text{drinks}) = \frac{1}{2}$$

$$P(\text{chocolate} \mid \text{eats}) = \frac{1}{1} \qquad P(\text{</s>} \mid \text{tea}) = \frac{1}{1}$$

Multiplying along <s> Lyn drinks chocolate </s> gives $\frac{2}{3} \times \frac{1}{2} \times \frac{1}{2} \times \frac{2}{2} = \frac{1}{6}$.

Quiz

Objective: Apply sequence probability approximation with bigrams after adding start and end word.

Question:
Given these conditional probabilities
P(Mary)=0.1; P(likes)=0.2; P(cats)=0.3
P(Mary|<s>)=0.2; P(</s>|cats)=0.6
P(likes|Mary) =0.3; P(cats|likes)=0.1

Approximate the probability of the following sentence with bigrams: "<s> Mary likes cats </s>"
Type: Multiple Choice, single answer

Options and solution:

1. P(<s> Mary likes cats </s>) = 0
2. P(<s> Mary likes cats </s>) =0.0036
3. P(<s> Mary likes cats </s>) = 0.003
4. P(<s> Mary likes cats </s>) = 1

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Count and Probability Matrices

Count Matrix: A matrix that counts occurrences of word pairs in a corpus.

Probability Matrix: A matrix that calculates probabilities of word pairs based on counts.

Example: For the sentence "I love NLP. I love AI."

- Count Matrix: Counts how many times each word appears with every other word.
- Probability Matrix: Calculates the probability of each word appearing given the previous word.

Count Matrix

$$P(w_n|w_{n-N+1}^{n-1}) = \frac{C(w_{n-N+1}^{n-1}, w_n)}{C(w_{n-N+1}^{n-1})}$$

- Rows: unique corpus (N-1)-grams
- Columns: unique corpus words

- Bigram count matrix

Corpus: <s>I study I learn</s>

	<s>	</s>	I	study	learn
<s>	0	0	1	0	0
</s>	0	0	0	0	0
I	0	0	0	1	1
study	0	0	1	0	0
learn	0	1	0	0	0

"study I" bigram

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Count Matrix Example

Text: "I love NLP. I love AI."

Bigrams:

- (I, love): 2
- (love, NLP): 1
- (love, AI): 1

Count Matrix:

- Rows: Words in the corpus
- Columns: Words in the corpus
- Cells: Count of occurrences of each word pair

Probability Matrix

$$P(w_n|w_{n-N+1}^{n-1}) = \frac{C(w_{n-N+1}^{n-1}, w_n)}{C(w_{n-N+1}^{n-1})}$$

- Divide each cell by its row sum

$$sum(row) = \sum_{w \in V} C(w_{n-N+1}^{n-1}, w) = C(w_{n-N+1}^{n-1})$$

Corpus: <s>I study I learn</s>

Count matrix (bigram)

	<s>	</s>	I	study	learn	sum
<s>	0	0	1	0	0	1
</s>	0	0	0	0	0	0
I	0	0	0	1	1	2
study	0	0	1	0	0	1
learn	0	1	0	0	0	1

Probability matrix

	<s>	</s>	I	study	learn
<s>	0	0	1	0	0
</s>	0	0	0	0	0
I	0	0	0	0.5	0.5
study	0	0	1	0	0
learn	0	1	0	0	0

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

Probability Matrix Example

Probability Calculation:

Bigram Probability:

$$P(\text{love} \mid \text{I}) = \frac{\text{Count}(\text{I}, \text{love})}{\text{Count}(\text{I})} = \frac{2}{2} = 1$$

Probability Matrix:

- Rows: Words in the corpus
- Columns: Words in the corpus
- Cells: Probability of each word given the previous word

From Probability Matrix to Language Model

A **language model** assigns a probability to any sequence of words. The probability matrix is one: read a row to predict the next word, multiply along a path to score a whole sentence.

- probability matrix => language model
  - Sentence probability
  - Next word prediction

	<s>	</s>	I	study	learn
<s>	0	0	1	0	0
</s>	0	0	0	0	0
I	0	0	0	0.5	0.5
study	0	0	1	0	0
learn	0	1	0	0	0

Sentence probability:
<s> I learn </s>

$$P(sentence) = P(I|<s>)P(learn|I)P(</s>|learn) = 1 \times 0.5 \times 1 = 0.5$$

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

N-gram Models: **Out of vocabulary words**

Out of Vocabulary (OOV) Words

Problem: Many words in a language are not present in the training corpus, leading to OOV issues.

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Closed vs. Open Vocabularies:

- **Closed vocabulary:** Only words seen during training are recognized.
- **Open vocabulary:** Model can handle unseen words.

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Closed vs. Open Vocabularies:

- **Closed vocabulary:** Only words seen during training are recognized.
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Solution: Use a special tag `<UNK>` in the corpus and input to represent unknown words.

Using <UNK> Tokens

Using <UNK>: Replace rare or unseen words with the special token <UNK>.

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Example:

- Original: `I love ChatGPT`
- With OOV handling: `I love <UNK>`

How to Use <UNK> in the Corpus

1. **Create vocabulary** V**:** Build a list of all words to be recognized (e.g., most frequent words).
2. **Replace OOV words:** For any word in the corpus not in V, replace it with <UNK>.
3. **Estimate probabilities:** Treat <UNK> as a regular word when computing word probabilities.

This approach allows the model to handle unseen words gracefully during inference.

Example

Corpus

<s> Lyn drinks chocolate </s>

<s> John drinks tea </s>

<s> Lyn eats chocolate </s>

Corpus

<s> Lyn drinks chocolate </s>

<s> <UNK> drinks <UNK> </s>

<s> Lyn <UNK> chocolate </s>

Min frequency f=2

Vocabulary

Lyn, drinks, chocolate

Input query

<s> Adam drinks chocolate</s>

<s><UNK> drinks chocolate</s>

Source: DeepLearning.AI, *NLP with Probabilistic Models*, Week 3 slides.

How to Create Vocabulary V

Criteria for Vocabulary Selection:

- **Minimum word frequency f:** Only include words that appear at least f times in the corpus.
- **Maximum vocabulary size $|V|$:** Limit V to the top $|V|$ most frequent words.

Use <UNK> Sparingly: Choose f and $|V|$ to minimize the number of words replaced by <UNK>, while keeping the vocabulary manageable.

Caution: A large <UNK> rate makes a model look good on its own terms, because predicting one frequent token is easy. Two models are only comparable if they were built with the same vocabulary V.

N-gram Models: **Smoothing**

Zero Counts Are Not Zero Probabilities

<UNK> fixes words missing from V. It does nothing for a pair of words that are both in V but never appeared *together*.

For the corpus `I am happy because I am learning` we computed

$$P(\text{happy} \mid \text{I}) = \frac{C(\text{I happy})}{C(\text{I})} = \frac{0}{2} = 0$$

even though "I" and "happy" are both perfectly ordinary words.

- One zero factor drives the whole sentence probability to 0.
- $\log 0 = -\infty$, so perplexity is undefined rather than merely bad.
- Longer contexts make this worse: most trigrams a model meets at test time were never seen in training.

Add-One (Laplace) Smoothing

Idea: pretend every possible n-gram was seen once more than it was.

$$P_{\text{Laplace}}(w_i \mid w_{i-1}) = \frac{C(w_{i-1}, w_i) + 1}{C(w_{i-1}) + |V|}$$

- The $+1$ in the numerator lifts unseen pairs off zero.
- The $+|V|$ in the denominator is what keeps each row summing to 1: one unit is added for every word that could follow w_{i-1}.
- **Add-k** is the same move with a smaller nudge, $k < 1$:

$$P_{\text{add-}k}(w_i \mid w_{i-1}) = \frac{C(w_{i-1}, w_i) + k}{C(w_{i-1}) + k\,|V|}$$

Add-One (Laplace) Smoothing (cont.)

Worked example. Same corpus, $V = \{\text{I, am, happy, because, learning}\}$, so $|V| = 5$ and $C(\text{I}) = 2$.

Unsmoothed

$$P(\text{am} \mid \text{I}) = \frac{2}{2} = 1$$

$$P(\text{happy} \mid \text{I}) = \frac{0}{2} = 0$$

Add-one

$$P(\text{am} \mid \text{I}) = \frac{2+1}{2+5} = \frac{3}{7}$$

$$P(\text{happy} \mid \text{I}) = \frac{0+1}{2+5} = \frac{1}{7}$$

Nothing is zero any more, and the five smoothed probabilities still sum to 1. The mass came out of the one bigram we actually observed, whose estimate fell from 1 to 3/7.

Add-One (Laplace) Smoothing (cont.)

The trade-off: that is a lot of mass to give away.

- Real vocabularies hold tens of thousands of words, so $|V|$ dwarfs $C(w_{i-1})$ and add-one hands almost all the probability to n-grams that were never seen.
- Add-k softens this but leaves the same flaw: every unseen n-gram gets the *same* estimate, whether it is plausible English or nonsense.
- Practical models instead fall back on shorter contexts, which are estimated from more data. If the trigram (w_{i-2}, w_{i-1}, w_i) is unseen, use the bigram (w_{i-1}, w_i); if that is unseen too, use the unigram.
  - **Backoff:** drop to the shorter context only when the longer one has count zero.
  - **Interpolation:** always mix all orders, $\hat{P} = \lambda_1 P(w_i) + \lambda_2 P(w_i \mid w_{i-1}) + \lambda_3 P(w_i \mid w_{i-2}, w_{i-1})$, with $\sum_j \lambda_j = 1$ and the λ_j tuned on held-out data.

Perplexity

- Perplexity (PPL) measures how well a language model predicts a sequence of words. Intuitively, it reflects "on average, how many choices the model is unsure about" per word.
- Formula:

$$\text{PPL}(W) = P(w_1, \ldots, w_N)^{-\frac{1}{N}} = \exp\left(-\frac{1}{N} \sum_{i=1}^{N} \log P(w_i \mid w_1, \ldots, w_{i-1}) \right)$$

- How to interpret:
  - Lower perplexity → model assigns higher probability to the test set sequences → better predictions.

Perplexity (cont.)

- Why it matters:
  - Standard metric for evaluating language models, from n-gram models to neural LMs.
  - Only compare perplexities of models that share the same vocabulary V.
- Example, scoring both models on the same held-out text with the same V:
  - Bigram model $\rightarrow$ PPL $\approx$ 200: on average it is choosing between roughly two hundred plausible next words.
  - Neural LM $\rightarrow$ PPL $\approx$ 20: the same choice, narrowed tenfold.
- Caveat: perplexity rewards fluency on held-out text, and nothing else. It says nothing about whether the model is truthful, useful, or safe.

N-gram Models: **Limitations**

N-gram Model Limitations

- **Data Sparsity**
- **Limited Context** (only N-1 words)
- **Explodes with Vocabulary Size**
- **Doesn't Capture Semantics/Syntax**

General NLP Challenges

Beyond N-grams, NLP as a whole still faces some big challenges:

- **Bias:** Models can pick up and even increase unfairness from the data they learn from.
- **Needs Lots of Data:** Modern NLP needs huge amounts of labeled text, which is hard and expensive to get.
- **Ambiguity:** Language can be confusing. In "I saw the man with the telescope", who has the telescope?
- **Many Languages:** It's tough to make models that work well for every language, especially those with little data.
- **Expensive to Train:** Training big models costs a lot of money and energy, so not everyone can do it.

N-gram Models: **Summary**

Beyond N-grams

- Word Embeddings (Word2Vec) and Neural Language Models (RNNs, LSTMs)
- Transformer-based Models (BERT, GPT)
- Subword Tokenization (Byte-Pair Encoding)
- Pretrained Language Models
- Contextual Representations

Summary

- NLP enables machines to understand human language
- N-grams model word sequences simply and effectively
- Sequence notation and probability estimation are essential
- Start/end/UNK tokens improve modeling and robustness
- Smoothing keeps unseen contexts from collapsing a sentence to probability zero
- Perplexity scores a model on held-out text, given a fixed vocabulary
- N-gram models are limited $\rightarrow$ neural models offer solutions

NLP: **References**

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